

Practical No. 8

Aim: Implementation of different types of functions with suitable examples.

Software Required: MySQL, Oracle 10g.

Theory:

In Database Management Systems, functions are predefined operations that simplify data processing and analysis. They allow users to perform tasks such as calculations, string manipulation, data conversion, and date handling efficiently within SQL queries. Types of functions are:

- Number Function
- Aggregate Function
- Character Function
- Conversion Function
- Date Function

Number Function:

Numeric Functions are used to perform operations on numeric data types such as INT, FLOAT, DECIMAL, DOUBLE, and others. These functions help perform operations like rounding, formatting, and aggregating data.

1. ABS(): The ABS() function returns the absolute value of a number, which is the number without its sign.

Syntax: SELECT ABS(number);

2. EXP(): The EXP() function returns the value of e raised to the power of a specified number, where e is the base of the natural logarithm.

Syntax: SELECT EXP(number);

3. POWER(): The POWER() function is used to raise a number to the power of another number.

Syntax: SELECT POWER(base, exponent);

4. MOD(): The MOD() function returns the remainder of a division operation.

Syntax: SELECT MOD(dividend, divisor);

5. ROUND(): The ROUND() function rounds a number to a specified number of decimal places.

Syntax: SELECT ROUND(number, decimal_places);

6. TRUNCATE(): The TRUNCATE() function is used to remove the decimal portion of a number without rounding. It truncates the number to the specified number of decimal places.

Syntax: SELECT TRUNCATE(number, decimal_places);

7. SQRT(): The SQRT() function returns the square root of a number.

Syntax: SELECT SQRT(number);

Aggregate Function:

These functions operate on multiple rows of a table, combining them into a single, more meaningful result.

1. COUNT: COUNT followed by a column name returns the count of tuple in that column. If DISTINCT keyword is used then it will return only the count of unique tuple in the column. Otherwise, it will return count of all the tuples.

Syntax: SELECT COUNT(*) AS alias_name FROM table_name;

2. SUM: SUM followed by a column name returns the sum of all the values in that column.

Syntax: SELECT SUM(column_name) AS alias_name FROM table_name;

3. AVG: AVG followed by a column name returns the average value of that column values.

Syntax: SELECT AVG(column_name) AS alias_name FROM table_name;

4. MAX: MAX followed by a column name returns the maximum value of that column.

Syntax: SELECT MAX(column_name) AS alias1 FROM table_name;

5. MIN: MIN followed by column name returns the minimum value of that column.

Syntax: SELECT MIN(column_name) AS alias1 FROM table_name;

Character function:

Character functions in SQL are used to manipulate, transform, and analyze string data. They help in changing case, measuring length, extracting substrings, searching within text, and modifying character values

1. UPPER: UPPER converts all characters in a string to uppercase letters.

Syntax: SELECT UPPER(column_name) AS alias_name FROM table_name;

2. LOWER: LOWER converts all characters in a string to lowercase letters.

Syntax: SELECT LOWER(column_name) AS alias_name FROM table_name;

3. LENGTH: LENGTH returns the number of characters present in a string.

Syntax: SELECT LENGTH(column_name) AS alias_name FROM table_name;

4. CONCAT: CONCAT joins two or more strings together into a single string.

Syntax: SELECT CONCAT(column1, column2) AS alias_name FROM table_name;

5. SUBSTRING: SUBSTRING extracts a portion of a string starting at a given position and for a specified length.

Syntax: SELECT SUBSTRING(column_name, start_position, length) AS alias_name FROM table_name;

6. INSTR: INSTR returns the numeric position of the first occurrence of a substring within a string.

Syntax: SELECT INSTR(column_name, 'substring') AS alias_name FROM table_name;

7. REPLACE: REPLACE substitutes all occurrences of a specified substring with another substring in a string.

Syntax: SELECT REPLACE(column_name, 'old_substring', 'new_substring') AS alias_name FROM table_name;

8. LPAD – Left-pads a string to a specified length with given characters.

Syntax: SELECT LPAD(column_name, length, pad_string) AS alias_name FROM table_name;

9. RPAD – Right-pads a string to a specified length with given characters.

Syntax: SELECT RPAD(column_name, length, pad_string) AS alias_name FROM table_name;

10. LTRIM – Removes specified leading characters from a string.

Syntax: SELECT TRIM(LEADING 'characters' FROM column_name) AS alias_name FROM table_name;

11. RTRIM – Removes specified trailing characters from a string.

Syntax: SELECT TRIM(TRAILING 'characters' FROM column_name) AS alias_name FROM table_name;

Conversion Functions:

Conversion functions in SQL are used to change data from one type to another, such as numbers to strings, strings to dates, or dates to formatted text.

1. CAST (to DECIMAL): CAST is used to convert a value from one data type to another, such as converting a string to a number.

Syntax: SELECT CAST(expression AS target_datatype) AS alias_name;

2. CAST (to CHAR): CAST can also convert numeric or other data types into character strings.

Syntax: SELECT CAST(column_name AS CHAR) AS alias_name FROM table_name;

3. DATE_FORMAT: DATE_FORMAT formats a date value into a string according to a specified format.

Syntax: SELECT DATE_FORMAT(date_column, 'format') AS alias_name FROM table_name;

4. STR_TO_DATE: STR_TO_DATE converts a string into a date by interpreting it using the given format mask.

Syntax: SELECT STR_TO_DATE('string', 'format') AS alias_name;

Date and Time Functions:

Date and Time functions in SQL are used to manipulate and extract information from date and time values. They help in formatting, calculating differences, adding or subtracting intervals, and comparing dates.

1. CURDATE: CURDATE returns the current date in the format YYYY-MM-DD.

Syntax: SELECT CURDATE() AS alias_name;

2. NOW: NOW returns the current system date and time.

Syntax: SELECT NOW() AS alias_name;

3. DATE_ADD: DATE_ADD adds a specified time interval (e.g., days, months, years) to a date.

Syntax: SELECT DATE_ADD(date_column, INTERVAL value unit) AS alias_name FROM table_name;

4. DATE_SUB: DATE_SUB subtracts a specified time interval (e.g., days, months, years) from a date.

Syntax: SELECT DATE_SUB(date_column, INTERVAL value unit) AS alias_name FROM table_name;

5. LAST_DAY: LAST_DAY returns the last day of the month for a given date.

Syntax: SELECT LAST_DAY(date_column) AS alias_name FROM table_name;

6. TIMESTAMPDIFF: TIMESTAMPDIFF returns the difference between two dates or timestamps in the specified unit (e.g., YEAR, MONTH, DAY).

Syntax: SELECT TIMESTAMPDIFF(unit, start_date, end_date) AS alias_name;

7. GREATEST: GREATEST returns the latest (maximum) date or the largest value among the given expressions.

Syntax: SELECT GREATEST(date1, date2) AS alias_name;

8. LEAST: LEAST returns the earliest (minimum) date or the smallest value among the given expressions.

Syntax: SELECT LEAST(date1, date2) AS alias_name;

Program:

```
CREATE DATABASE functions_demo;
```

```
USE functions_demo;
```

```
CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50) NOT NULL, dept VARCHAR(50), salary DECIMAL(10,2), bonus DECIMAL(10,2), hire_date DATE, city VARCHAR(50));
```

```
INSERT INTO employees VALUES
```

```
(1, 'Elon Musk', 'Management', 95000.75, 5000.25, '2020-01-15', 'New York'),  
(2, 'Larry Page', 'R&D', 88000.50, 4000.50, '2019-03-22', 'San Francisco'),  
(3, 'Sundar Pichai', 'R&D', 99000.00, 6000.75, '2021-07-05', 'Los Angeles'),  
(4, 'Satya Nadella', 'Sales', 75000.25, 3000.25, '2018-11-30', 'Chicago'),  
(5, 'Tim Cook', 'Management', 105000.00, 7000.00, '2022-06-01', 'Boston');
```

```
-- Number Functions
```

```
SELECT ABS(-25) AS AbsoluteValue;  
SELECT POWER(salary, 2) AS SalarySquared FROM employees;  
SELECT MOD(emp_id, 2) AS EvenOdd FROM employees;  
SELECT ROUND(bonus, 1) AS RoundedBonus FROM employees;  
SELECT TRUNCATE(salary, 0) AS TruncatedSalary FROM employees;  
SELECT SQRT(81) AS SquareRoot;
```

```
-- Aggregate Functions
```

```
SELECT COUNT(*) AS TotalEmployees FROM employees;  
SELECT SUM(salary) AS TotalSalary FROM employees;  
SELECT AVG(salary) AS AverageSalary FROM employees;  
SELECT MIN(salary) AS MinSalary, MAX(salary) AS MaxSalary FROM employees;  
SELECT dept, AVG(salary) AS AvgDeptSalary FROM employees GROUP BY dept;
```

```
-- String/Character Functions
```

```
SELECT UPPER(emp_name) AS UpperName FROM employees;  
SELECT LOWER(emp_name) AS LowerName FROM employees;  
SELECT LENGTH(emp_name) AS NameLength FROM employees;  
SELECT CONCAT(emp_name, ' - ', dept) AS EmployeeDept FROM employees;  
SELECT SUBSTR(emp_name, 1, 5) AS ShortName FROM employees;  
SELECT INSTR(emp_name, 'a') AS PositionOfA FROM employees;  
SELECT REPLACE(city, 'o', '0') AS CityWithReplacedO FROM employees;  
SELECT LPAD(emp_name, 15, '*') AS LpadName FROM employees;  
SELECT RPAD(emp_name, 15, '*') AS RpadName FROM employees;  
SELECT TRIM(LEADING 'S' FROM 'SSMITHSS') AS LtrimExample;  
SELECT TRIM(TRAILING 'S' FROM 'SSMITHSS') AS RtrimExample;
```

```
-- Conversion Functions
```

```
SELECT CAST('1234.56' AS DECIMAL(10,2)) AS ToNumber;  
SELECT CAST(salary AS CHAR) AS SalaryAsText FROM employees;
```

```
SELECT DATE_FORMAT(hire_date, '%d-%M-%Y') AS FormattedHireDate FROM employees;
SELECT STR_TO_DATE('15-02-2021', '%d-%m-%Y') AS ToDate;
```

-- Date Functions

```
SELECT CURDATE() AS Today;
SELECT NOW() AS CurrentDateTime;
SELECT DATE_ADD(hire_date, INTERVAL 6 MONTH) AS SixMonthsAfterHire FROM employees;
SELECT DATE_SUB(hire_date, INTERVAL 1 YEAR) AS OneYearBeforeHire FROM employees;
SELECT LAST_DAY(hire_date) AS LastDayOfHireMonth FROM employees;
SELECT TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) AS YearsWorked FROM employees;
SELECT GREATEST(hire_date, '2020-01-01') AS GreatestDate FROM employees;
SELECT LEAST(hire_date, '2020-01-01') AS LeastDate FROM employees;
```

Output:

```
mysql> SELECT CURDATE() AS Today;
+-----+
| Today |
+-----+
| 2025-09-30 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT NOW() AS CurrentDateTime;
+-----+
| CurrentDateTime |
+-----+
| 2025-09-30 22:31:19 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT DATE_ADD(hire_date, INTERVAL 6 MONTH) AS SixMonthsAfterHire FROM employees;
+-----+
| SixMonthsAfterHire |
+-----+
| 2020-07-15 |
| 2019-09-22 |
| 2022-01-05 |
| 2019-05-30 |
| 2022-12-01 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT DATE_SUB(hire_date, INTERVAL 1 YEAR) AS OneYearBeforeHire FROM employees;
+-----+
| OneYearBeforeHire |
+-----+
| 2019-01-15 |
| 2018-03-22 |
| 2020-07-05 |
| 2017-11-30 |
| 2021-06-01 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LAST_DAY(hire_date) AS LastDayOfHireMonth FROM employees;
+-----+
| LastDayOfHireMonth |
+-----+
| 2020-01-31 |
| 2019-01-31 |
| 2021-07-31 |
| 2018-11-30 |
| 2022-06-30 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT CAST('1234.56' AS DECIMAL(10,2)) AS ToNumber;
+-----+
| ToNumber |
+-----+
| 1234.56 |
+-----+
1 row in set (0.03 sec)

mysql> SELECT CAST(salary AS CHAR) AS SalaryAsText FROM employees;
+-----+
| SalaryAsText |
+-----+
| 95000.75 |
| 88000.50 |
| 99000.00 |
| 75000.25 |
| 105000.00 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT DATE_FORMAT(hire_date, '%d-%M-%Y') AS FormattedHireDate FROM employees;
+-----+
| FormattedHireDate |
+-----+
| 15-January-2020 |
| 22-March-2019 |
| 05-July-2021 |
| 30-November-2018 |
| 01-June-2022 |
+-----+
5 rows in set (0.01 sec)

mysql> SELECT STR_TO_DATE('15-02-2021', '%d-%m-%Y') AS ToDate;
+-----+
| ToDate |
+-----+
| 2021-02-15 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) AS YearsWorked FROM employees;
+-----+
| YearsWorked |
+-----+
| 5 |
| 6 |
| 4 |
| 6 |
| 3 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT GREATEST(hire_date, '2020-01-01') AS GreatestDate FROM employees;
+-----+
| GreatestDate |
+-----+
| 2020-01-15 |
| 2020-01-01 |
| 2021-07-05 |
| 2020-01-01 |
| 2022-06-01 |
+-----+
5 rows in set (0.03 sec)

mysql> SELECT LEAST(hire_date, '2020-01-01') AS LeastDate FROM employees;
+-----+
| LeastDate |
+-----+
| 2020-01-01 |
| 2019-03-22 |
| 2020-01-01 |
| 2018-11-30 |
| 2020-01-01 |
+-----+
5 rows in set (0.00 sec)
```

```

mysql> -- Number Functions
mysql> SELECT ABS(-25) AS AbsoluteValue;
+-----+
| AbsoluteValue |
+-----+
|          25 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT POWER(salary, 2) AS SalarySquared FROM employees;
+-----+
| SalarySquared |
+-----+
| 9025142500.5625 |
| 774088000.25 |
| 9001000000 |
| 5625037500.0625 |
| 11025000000 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT MOD(emp_id, 2) AS EvenOdd FROM employees;
+-----+
| EvenOdd |
+-----+
| 1 |
| 0 |
| 1 |
| 0 |
| 1 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT ROUND(bonus, 1) AS RoundedBonus FROM employees;
+-----+
| RoundedBonus |
+-----+
| 5000.3 |
| 4000.5 |
| 6000.8 |
| 3000.3 |
| 7000.6 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT TRUNCATE(salary, 0) AS TruncatedSalary FROM employees;
+-----+
| TruncatedSalary |
+-----+
| 95000 |
| 88000 |
| 99000 |
| 75000 |
| 105000 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT SQRT(81) AS SquareRoot;
+-----+
| SquareRoot |
+-----+
| 9 |
+-----+
1 row in set (0.00 sec)

```

```

mysql> -- Aggregate Functions
mysql> SELECT COUNT(*) AS TotalEmployees FROM employees;
+-----+
| TotalEmployees |
+-----+
| 5 |
+-----+
1 row in set (0.04 sec)

mysql> SELECT SUM(salary) AS TotalSalary FROM employees;
+-----+
| TotalSalary |
+-----+
| 462001.50 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(salary) AS AverageSalary FROM employees;
+-----+
| AverageSalary |
+-----+
| 92400.300000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MIN(salary) AS MinSalary, MAX(salary) AS MaxSalary FROM employees;
+-----+
| MinSalary | MaxSalary |
+-----+
| 75000.25 | 105000.00 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT dept, AVG(salary) AS AvgDeptSalary FROM employees GROUP BY dept;
+-----+
| dept | AvgDeptSalary |
+-----+
| Management | 100000.375000 |
| R&D | 93500.250000 |
| Sales | 75000.250000 |
+-----+
3 rows in set (0.00 sec)

```

```

mysql> -- String Character Functions
mysql> SELECT UPPER(emp_name) AS UpperName FROM employees;
+-----+
| UpperName |
+-----+
| ELON MUSK |
| LARRY PAGE |
| SUNDAR PICHAI |
| SATYA NADELLA |
| TIM COOK |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LOWER(emp_name) AS LowerName FROM employees;
+-----+
| LowerName |
+-----+
| elon musk |
| larry page |
| sundar pichai |
| satya nadella |
| tim cook |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LENGTH(emp_name) AS NameLength FROM employees;
+-----+
| NameLength |
+-----+
| 9 |
| 10 |
| 13 |
| 13 |
| 8 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT CONCAT(emp_name, ' - ', dept) AS EmployeeDept FROM employees;
+-----+
| EmployeeDept |
+-----+
| Elon Musk - Management |
| Larry Page - R&D |
| Sundar Pichai - R&D |
| Satya Nadella - Sales |
| Tim Cook - Management |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT SUBSTR(emp_name, 1, 5) AS ShortName FROM employees;
+-----+
| ShortName |
+-----+
| Elon |
| Larry |
| Sunda |
| Satya |
| Tim C |
+-----+
5 rows in set (0.00 sec)

```

```

mysql> SELECT INSTR(emp_name, 'a') AS PositionOfA FROM employees;
+-----+
| PositionOfA |
+-----+
| 0 |
| 2 |
| 5 |
| 2 |
| 0 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT REPLACE(city, 'o', '0') AS CityWithReplacedO FROM employees;
+-----+
| CityWithReplacedO |
+-----+
| New Y0rk |
| San Francisco0 |
| L0s Angeles |
| Chicag0 |
| B0ston |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LPAD(emp_name, 15, '*') AS LpadName FROM employees;
+-----+
| LpadName |
+-----+
| *****Elon Musk |
| *****Larry Page |
| *****Sundar Pichai |
| *****Satya Nadella |
| *****Tim Cook |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT RPAD(emp_name, 15, '*') AS RpadName FROM employees;
+-----+
| RpadName |
+-----+
| Elon Musk***** |
| Larry Page***** |
| Sundar Pichai** |
| Satya Nadella** |
| Tim Cook***** |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT TRIM(LEADING 'S' FROM 'SSMITHSS') AS LtrimExample;
+-----+
| LtrimExample |
+-----+
| MITHSS |
+-----+
1 row in set (0.03 sec)

mysql> SELECT TRIM(TRAILING 'S' FROM 'SSMITHSS') AS RtrimExample;
+-----+
| RtrimExample |
+-----+
| SSMITH |
+-----+
1 row in set (0.00 sec)

```

Conclusion: Hence, we have successfully studied and implemented different types of functions with suitable examples.